

Semantics-Aware Explainable Machine Learning for Urban Service Demand Forecasting in Smart Cities

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Abstract

Urban service platforms such as 311 systems create large-scale traces of reported municipal demand, but forecasting actionable demand surges remains difficult because requests are spatially uneven, semantically heterogeneous, temporally non-stationary, and affected by reporting behavior. This paper formulates reported urban service-demand forecasting as a spatiotemporal-semantic early-warning task at the neighborhood-week-complaint-category level. Using New York City as a case study, we integrate more than 30.9 million geocoded 311 requests with historical demand dynamics, calendar and COVID-period indicators, NOAA weather exposure, OpenStreetMap point-of-interest context, and PLUTO land-use and built-environment features. The target is a next-week abnormal increase in reported service demand, defined from shifted historical rolling statistics so that forecasting features do not leak future information. Models are evaluated with a target-week chronological protocol: target weeks in 2015–2022 for training, 2023 target weeks for validation, and 2024–2025 target weeks for held-out future testing; validation and test positive shares are approximately 12.7%. The selected decision strategy averages LightGBM and XGBoost probability scores and applies complaint-category-specific thresholds selected only on the validation period. Under this target-week chronological evaluation, it achieves future-test $F1 = 0.3800$, precision = 0.3012, recall = 0.5146, PR-AUC = 0.3294, and ROC-AUC = 0.7661. SHAP analysis of the LightGBM component shows that historical temporal demand dynamics dominate prediction, while built-environment, POI, weather, semantic-category, and calendar features provide complementary signals. The results support a decision-support

interpretation: the model identifies neighborhoods and service categories with elevated risk of abnormal reported demand, while recognizing that 311 data capture reported service demand rather than all underlying urban conditions.

Keywords: urban computing, smart cities, 311 service requests, service-demand forecasting, explainable machine learning, SHAP, spatiotemporal modeling

1 Introduction

Smart-city service systems increasingly rely on administrative data streams to understand where and when public needs emerge. Among these streams, 311 requests are valuable because they record resident-facing reports across municipal service domains. Prior work shows that 311 data contain spatial signatures of urban conditions [1], while also reflecting uneven participation and reporting behavior [2, 3]. Thus, 311 data can support city operations and research, but should not be treated as a complete census of all urban problems.

This paper studies NYC 311 requests as a large-scale case study in reported urban service-demand forecasting. Rather than predicting raw complaint counts, we formulate an early-warning task: for each neighborhood, week, and semantic complaint category, estimate whether the following week will exhibit an abnormal increase in reported demand. This aligns with operations, where analysts need near-term category-specific signals rather than one citywide count.

We build a multi-source, leakage-safe, and explainable pipeline over a dense NTA-week-complaint-category panel. Historical 311 demand is combined with calendar timing, COVID-period indicators, weekly NOAA weather summaries, OSM POI functional context, and PLUTO land-use and built-environment features. Because the target is a next-week abnormal increase, all lag and rolling-window features are shifted so that only pre-prediction information is used.

The empirical evaluation follows a target-week chronological design. Models are trained using target weeks in 2015–2022, tuned using 2023 target weeks, and evaluated on a held-out future test period covering 2024–2025 target weeks. This split is central because random splits would mix future and past observations from the same neighborhood-category series and overstate forecasting validity. Since abnormal increases are relatively rare, the evaluation emphasizes F1, precision, recall, PR-AUC, ROC-AUC, balanced accuracy, and confusion matrices rather than raw accuracy.

The final selected strategy averages tuned LightGBM and XGBoost scores and applies complaint-category-specific thresholds selected on validation. It is a decision-support layer, not an automated allocation system: it ranks potential abnormal demand increases while making false positives and false negatives visible. We explain a tuned LightGBM component using SHAP to identify whether predictions are driven by historical dynamics, weather, semantic category, POI context, land use, or calendar effects.

The study asks three research questions: (RQ1) can a leakage-safe multi-source panel forecast next-week abnormal reported demand under a future chronological test;

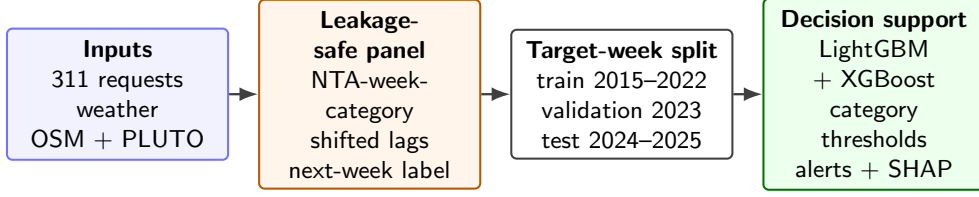


Fig. 1 End-to-end leakage-safe workflow. Features predict the next target week, chronological periods are split by target week before fitting, and the final layer reports alerts, metrics, and SHAP drivers for reported demand rather than causal effects.

(RQ2) do semantic service categories benefit from category-specific alert thresholds; and (RQ3) which temporal and urban-context feature groups drive the predictions? The contributions are: (1) a spatiotemporal-semantic early-warning formulation; (2) a large NYC panel built from 30,940,129 NTA-matched 311 requests and multi-source urban context; (3) a target-week chronological evaluation with validation-only threshold selection, baselines, and feature ablations; and (4) SHAP explanations that distinguish predictive associations from causal claims. Figure 1 summarizes this workflow.

2 Related Work

2.1 Urban computing and citizen-generated service data

Urban computing integrates sensing, administrative, and geographic data to understand city dynamics and support decision making [4]. In this setting, 311 service requests link residents, reported concerns, and municipal response systems. Prior work shows that 311 request structure can serve as a signature of urban location [1], but citizen-generated data also reflect awareness, trust, access, reporting propensity, and governance processes [2, 3]. These findings motivate the central caution in this paper: we forecast abnormal reported service demand, not all true urban problems. Domain studies such as NYC flooding analysis through 311 complaints [5] further show why weather and service reports can be linked, while our contribution is a citywide multi-category next-week warning task.

2.2 Urban context from POIs and land use

Neighborhood function shapes reported service demand. OSM is a major volunteered geographic information source [6]; OSM POIs have been used to model urban change [7], and POI research highlights both functional value and quality-control challenges [8]. Land-use and built-environment data provide complementary stable context. Open big data can map essential urban land-use categories [9], and functional-zone research emphasizes combining POIs, remote sensing, and geospatial data [10]. We therefore use aggregate POI, diversity, and PLUTO features as contextual signals rather than as exact measurements of demand.

2.3 Imbalanced forecasting, gradient boosting, and explainability

The target in this paper is an abnormal increase event. This connects to anomaly-detection literature [11], but with a supervised and operationally defined next-week label. Since positive events are much less frequent than normal observations, PR-AUC and F1 are more informative than accuracy alone. Saito and Rehmsmeier [12] show why precision-recall evaluation is especially useful for imbalanced classification.

Gradient-boosted tree methods are strong tabular-learning baselines and are well suited to nonlinear interactions between temporal and contextual features. The modeling stack in this study builds on the random forest and gradient boosting traditions [13, 14], and uses XGBoost [15] and LightGBM [16] as the main tuned models. Because operational use of such models requires interpretability, we apply SHAP [17] and TreeSHAP [18] to explain a tuned LightGBM component. We follow general XAI cautions [19, 20]: explanation values describe model behavior and should not be interpreted as causal effects.

3 Data and Task Formulation

3.1 Data sources and analytical unit

The project integrates five evidence sources from official public data portals¹: NYC Open Data 311 Service Requests from 2015–2025 spatially joined to 2020 NTAs; nine stable semantic complaint categories; NOAA Global Historical Climatology Network Daily weather aggregated to weekly exposure; OpenStreetMap POIs aggregated to NTA-level counts, density, entropy, and mixed-use indicators; and NYC Planning PLUTO/MapPLUTO land-use and built-environment summaries.

The resulting analytical unit is

$$(n, t, c),$$

where n denotes an NTA, t a week start date, and c a semantic complaint category. The dense panel contains 262 NTAs, 575 weekly periods, and nine complaint categories, yielding 1,355,850 rows. Table 1 summarizes the most important construction facts.

3.2 Semantic complaint categories

NYC 311 complaint types are highly granular and can change over time. Modeling each raw type separately would create sparse series, while collapsing all requests would erase service meaning. We therefore use stable semantic categories: noise, housing, water and sewer, public safety, parking and traffic, environmental issues, infrastructure, sanitation, and a residual other category retained to avoid false precision. Representative raw types include Noise–Residential and Noise–Street/Sidewalk for noise, Heat/Hot

¹Official source pages: NYC Open Data 311, NOAA GHCN-Daily, OpenStreetMap copyright/license, and NYC Planning PLUTO/MapPLUTO.

Table 1 Dataset construction summary.

Item	Value
NTA-matched 311 requests	30,940,129
Spatial match rate after cleaning	99.9777%
Analytical unit	NTA $\times$ week $\times$ complaint category
Dense panel rows	1,355,850
NTAs	262
Weeks	575
Complaint categories	9
Temporal coverage	2014-12-29 to 2025-12-29 weekly panel
NOAA daily observations	4,018 daily rows, aggregated to 575 weeks
OSM POIs after NTA assignment	77,083
Final output columns	252
Model-ready selected features	215

Water and Plumbing for housing, Illegal Parking and Blocked Driveway for parking/traffic, Street Condition and Sidewalk Condition for infrastructure, and Missed Collection or Dirty Condition for sanitation.

3.3 Next-week abnormal-increase target

For each neighborhood n , week t , and complaint category c , the feature vector is

$$x_{n,t,c} = [\text{historical demand, calendar, weather, OSM POI context, PLUTO context}].$$

Here t is the feature week: all predictors are available after that week closes. The target week is $t + 1$, seven days after ‘week_start’. The supervised target indicates whether this following week has an abnormal increase in reported service demand:

$$y_{n,t,c} = \mathbb{I}\{\text{count}_{n,t+1,c} > \mu_{n,t,c}^{8w} + 1.5\sigma_{n,t,c}^{8w}\},$$

where $\mu_{n,t,c}^{8w}$ and $\sigma_{n,t,c}^{8w}$ are computed from a shifted historical eight-week window. The shift ensures that rolling statistics and lag features use only information available before the prediction week. The target focuses on future surges relative to each local historical baseline rather than absolute volume alone.

4 Methodology

4.1 Feature groups

The model-ready configuration contains 215 selected features. Historical demand features include lag counts, rolling statistics, changes, ratios to historical baselines, current complaint count, and history-availability flags. Calendar and period features include seasonal timing and COVID-period indicators. Weather features summarize

weekly temperature, precipitation, snow, wind, and extreme-day counts. OSM POI features describe functional context through semantic counts, densities, shares, entropy, diversity, and tag-family summaries. PLUTO features summarize built-environment and land-use composition, including lots, building area, residential and commercial area, land-use ratios, and parcel density.

4.2 Models

The pipeline compares simple baselines, logistic regression, gradient-boosted trees, and tuned LightGBM/XGBoost variants. Tuned gradient-boosted models provide the strongest validation performance and capture nonlinear tabular interactions. Hyperparameter tuning searches complexity, learning rate, regularization, and positive-class weighting. For reproducible final model selection under local runtime constraints, the validation-ranked LightGBM and XGBoost candidates used in the final decision comparison are trained on a stratified 300,000-row sample from the target-week training period, preserving the positive-class proportion. Scores are converted into hard alerts using thresholds selected only on validation.

4.3 Semantics-aware ensemble decision layer

Let $s_{n,t,c}^L$ be the tuned LightGBM score and $s_{n,t,c}^X$ the tuned XGBoost score. The ensemble score is

$$s_{n,t,c}^E = 0.5s_{n,t,c}^L + 0.5s_{n,t,c}^X.$$

We compare global and complaint-category-specific thresholds. In the category-specific strategy, each category c receives a threshold τ_c selected on the 2023 validation period to optimize F1. The final alert is

$$\hat{y}_{n,t,c} = \mathbb{I}\{s_{n,t,c}^E \geq \tau_c\}.$$

This is semantics-aware at the decision layer: the same score can imply different alert decisions because score distributions and operational trade-offs differ by service domain.

4.4 Explainability

We use SHAP to explain the tuned LightGBM component. The final decision strategy is an ensemble with category-specific thresholds, so SHAP is interpreted as evidence about a representative tree component trained under the same feature design and chronological protocol, not as a direct explanation of every ensemble alert or as causal attribution.

5 Experimental Setup

5.1 Chronological split

The main experiment uses a chronological split by target week, not by feature week: target weeks in 2015–2022 are used for training, 2023 target weeks for validation, and

Table 2 Compact baseline and feature-ablation ranking checks on the 2024–2025 target-week test period. The table reports threshold-independent score quality; final validation-selected hard-alert performance is reported in Table 3.

Evidence	Model/setting	PR-AUC	ROC-AUC
Baseline	Current-week persistence	0.2359	0.6389
Baseline	Logistic, historical only	0.2582	0.6943
Ablation	LightGBM, historical only	0.2770	0.7261
Ablation	LightGBM, full context	0.3158	0.7549

2024–2025 target weeks for testing. This means, for example, that a row whose feature week ends in late 2022 but predicts an early-2023 target week is assigned to validation rather than training. Rows without an observable next-week target are excluded from model-ready evaluation. This target-week assignment is applied before model fitting, validation ranking, threshold tuning, and test reporting.

The model-ready target-week split contains 954,990 train rows with 121,306 positives, 122,616 validation rows with 15,574 positives, and 247,590 test rows with 31,570 positives. The positive class is rare but stable, with approximately 12.7% positive share in validation and test. This imbalance motivates evaluation with precision, recall, F1, PR-AUC, ROC-AUC, balanced accuracy, and confusion matrices.

5.2 Metrics and model selection

F1 is used as the primary validation-ranking metric because the task requires balancing alert precision and event detection. Precision measures how many alerts correspond to true abnormal increases, while recall measures how many true abnormal increases are captured. PR-AUC evaluates ranking quality under class imbalance, and ROC-AUC is reported as a complementary threshold-independent metric. Balanced accuracy is included to summarize sensitivity and specificity without letting the majority class dominate.

All reported thresholds are selected on the validation period only. Threshold tuning is treated as a method component rather than a cosmetic post-processing step because the default 0.5 threshold is poorly matched to an imbalanced early-warning task. Importantly, threshold tuning changes hard classification metrics such as F1, precision, recall, and predicted-positive share; it does not change the underlying PR-AUC or ROC-AUC for a fixed score vector.

6 Results

6.1 Baselines, ablations, and final model comparison

Table 2 gives compact baseline and feature-ablation evidence under the target-week split. It reports only ranking metrics, so the comparison is not affected by the mixed hard-threshold protocols used by the underlying checks.

Table 2 indicates that learned models and full-context features improve ranking quality, particularly PR-AUC. Final hard-alert performance additionally depends

Table 3 Final decision-layer comparison. Thresholds are selected on validation and evaluated unchanged on the future test period.

Model	Threshold	Val F1	Val PR-AUC	Test F1	Test Precision	Test Recall	Test PR-AUC
Ensemble LGBM(0.50)+XGB(0.50)	Per category	0.3756	0.3137	0.3800	0.3012	0.5146	0.3294
Tuned LightGBM	Per category	0.3744	0.3089	0.3789	0.3000	0.5140	0.3270
Ensemble LGBM(0.50)+XGB(0.50)	Global	0.3720	0.3137	0.3797	0.2967	0.5269	0.3294
Tuned XGBoost	Per category	0.3704	0.3119	0.3761	0.2901	0.5347	0.3247

Table 4 Precision and lift for top-ranked risk sets on the 2024–2025 test period. Top-risk rows are selected by score rank; all alerts are selected by the threshold layer.

Risk set	Rows / selected cases	Precision	Recall	Lift
Top 1% risk	2,476	0.5727	0.0449	4.49
Top 5% risk	12,380	0.4426	0.1736	3.47
Top 10% risk	24,759	0.3782	0.2966	2.97
All alerts	53,941	0.3012	0.5146	2.36

on model tuning, ensemble averaging, and validation-selected thresholds, which are compared in Table 3.

Table 3 compares the final decision strategies. The selected model is the ensemble of LightGBM and XGBoost with complaint-category-specific thresholds. On the held-out 2024–2025 target-week test period, it achieves $F1 = 0.3800$, precision = 0.3012, recall = 0.5146, PR-AUC = 0.3294, ROC-AUC = 0.7661, and balanced accuracy = 0.6701.

6.2 Alert interpretation

The final test confusion matrix contains 16,247 true positives, 37,694 false positives, 15,323 false negatives, and 178,326 true negatives. This means the model captures about half of true abnormal demand increases while roughly three out of ten alerts correspond to true abnormal increases. Such performance is realistic for an imbalanced reported-demand early-warning task and is most useful as a prioritization layer.

Because the operational use case is prioritization, Table 4 evaluates concentration of true events among the highest-risk predictions. The top 1% of scored test rows has precision 0.5727, corresponding to 4.49 times the test base rate, while the all-alert decision layer retains recall 0.5146 at 2.36 times the base rate.

Compared with logistic-regression baselines, gradient-boosted models improve ranking quality and alert conversion: the best listed logistic-regression candidate obtains PR-AUC = 0.2582, whereas the selected ensemble obtains PR-AUC = 0.3294 and $F1 = 0.3800$ under the final validation-selected thresholds.

6.3 Performance by complaint category

Table 5 shows that performance varies across municipal service domains. Noise has the strongest test F1, followed by housing, public safety, and water/sewer. Sanitation, infrastructure, and environment are weaker by F1, although several retain moderate

Table 5 Category-specific thresholds and test performance. Base and alert rates are measured on the 2024–2025 target-week test period.

Category	Threshold	Base rate	Alert rate	F1	Precision	Recall
Noise	0.1700	0.1447	0.2593	0.4423	0.3446	0.6173
Housing	0.1901	0.1352	0.1887	0.4049	0.3476	0.4849
Public safety	0.1571	0.1285	0.2174	0.3968	0.3157	0.5339
Water/sewer	0.2050	0.1258	0.1865	0.3964	0.3319	0.4919
Other	0.1550	0.1316	0.2486	0.3725	0.2849	0.5381
Parking/traffic	0.1650	0.1304	0.2403	0.3644	0.2810	0.5181
Environment	0.1576	0.1053	0.2163	0.3585	0.2665	0.5477
Infrastructure	0.1879	0.1307	0.2248	0.3495	0.2764	0.4753
Sanitation	0.1773	0.1153	0.1790	0.3194	0.2625	0.4076

recall. This heterogeneity supports the paper’s central argument that urban service-demand forecasting is not a single homogeneous time-series problem.

7 Explainability Analysis

7.1 Global feature importance

SHAP analysis is performed on the tuned LightGBM component. Figure 2 summarizes SHAP importance by feature group. Historical temporal features dominate, accounting for 58.82% of summed mean absolute SHAP importance. This is expected: recent volatility, ratios to historical means, seasonal recurrence, and lag counts directly encode demand dynamics. However, contextual features are not negligible. PLUTO built-environment features account for 11.49%, OSM POI context for 10.09%, weather for 6.06%, semantic category for 6.01%, and calendar features for 4.60%.

The top individual features are also consistent with the abnormal-increase formulation. The leading features include ‘rolling_8w_std’, ‘ratio_to_8w_mean’, ‘complaint_category’, ‘lag_52w_count’, ‘week_of_year’, ‘rolling_4w_mean’, ‘rolling_12w_std’, and current ‘complaint_count’. These features combine short-term volatility, deviation from recent baseline, annual seasonality, service semantics, and current observed demand.

7.2 Category-specific interpretation

Category-specific SHAP summaries indicate that the highest-ranked features are broadly similar across categories, but their relative magnitudes differ. In every category, the top features include rolling eight-week standard deviation, ratio to the eight-week mean, complaint category, 52-week lag count, and week of year. This pattern supports two interpretations. First, the model is primarily learning temporal abnormality relative to local historical behavior. Second, semantic category remains important even after category-specific series are included, suggesting that service domains differ in baseline volatility, seasonality, and alertability.

Two local SHAP examples further show how the explanation layer should be used. For target week 2025-10-13, a true-positive housing alert in Hamilton Heights–Sugar

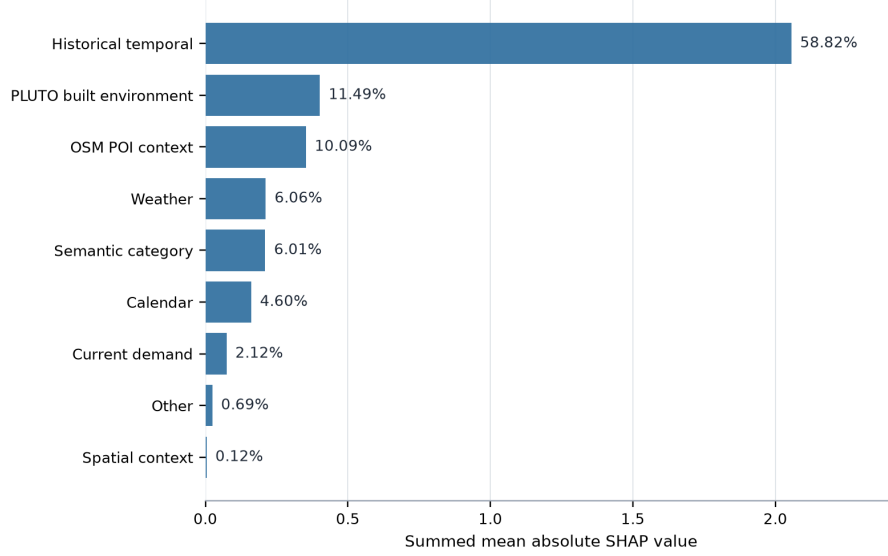


Fig. 2 Feature-group SHAP importance for the tuned LightGBM explanation backbone. Historical temporal demand features dominate, while PLUTO, OSM POI, weather, semantic-category, and calendar features provide complementary predictive information.

Hill (score 0.903, predicted alert, true label 1) is pushed upward by history length, 52-week lag demand, and residential-lot features. For the same target week, a false-positive housing alert in Morrisania (score 0.834, predicted alert, true label 0) is pushed upward by category, ratio to the eight-week mean, one-week lag demand, the four-week rolling mean, and current complaint count. These examples illustrate why explanations should support analyst review rather than automatic action.

The built-environment and POI contributions should be interpreted as contextual associations. PLUTO density or residential-area features may help distinguish neighborhoods with different housing-related demand dynamics, while POI entropy and named POI counts may reflect functional intensity or mixed-use conditions. These explanations do not prove that changing a POI or land-use feature would causally change 311 demand; they describe how the trained model uses observed context to estimate reported-demand risk.

8 Discussion

8.1 Operational meaning

The final model is best understood as early-warning decision support. On the future test period, recall of 0.5146 means it captures slightly more than half of true abnormal increases, while precision of 0.3012 means many alerts are false positives. In practice, the system should be used as a triage layer that narrows analyst attention to high-risk NTA-category weeks, with final decisions incorporating agency capacity, complaint severity, and local knowledge.

Complaint-category-specific thresholds make this interpretation more practical. A single global threshold assumes all domains share the same alert rule, but noise, housing, infrastructure, sanitation, and public-safety requests differ in baseline frequency, volatility, reporting behavior, and response costs. A semantics-aware threshold layer reflects these differences while retaining a common modeling framework.

8.2 Methodological implications

The results reinforce three methodological points. First, temporal leakage control is essential; random splits are poorly aligned with prospective forecasting in repeated neighborhood-category panels. Second, threshold selection should be reported transparently in imbalanced early-warning tasks because the default 0.5 threshold can produce impractical alert rates. Third, context features should be evaluated as complements to, not replacements for, historical demand. SHAP shows that temporal dynamics dominate, but neighborhood function, built environment, weather, category semantics, and calendar timing still contribute information.

8.3 Limitations

Several limitations should guide interpretation. First, 311 data capture reported service demand, not all true urban problems, and reporting propensity may reproduce socio-spatial inequities. Second, OSM POI and PLUTO features are used as relatively stable neighborhood context and may not capture within-period changes over 2015–2025. Third, NOAA weather is aggregated as weekly city-level exposure and may not capture microclimatic variation across NTAs. Fourth, tuned LightGBM and XGBoost runs use stratified sampled training for local computational reproducibility; future work should verify robustness under repeated full-training runs. Fifth, the target depends on an eight-week historical baseline and a 1.5 standard-deviation threshold; alternative definitions may produce different alert behavior. Sixth, the model is evaluated in NYC only. Finally, SHAP explains predictive behavior, not causal effects of weather, POI density, or land-use characteristics.

9 Conclusion

This paper presented a semantics-aware explainable machine-learning pipeline for forecasting abnormal reported urban service-demand increases in NYC 311 data. By modeling the NTA-week-complaint-category panel, integrating historical demand with weather, OSM POI, and PLUTO context, and evaluating with a target-week chronological split, the study frames 311 forecasting as a realistic early-warning task rather than a generic count-prediction exercise. Under the target-week chronological evaluation, the selected ensemble of LightGBM and XGBoost with complaint-category-specific thresholds achieved future-test $F1 = 0.3800$, precision = 0.3012, recall = 0.5146, PR-AUC = 0.3294, and ROC-AUC = 0.7661. SHAP analysis showed that historical temporal dynamics are the dominant predictive signal, while built environment, POI context, weather, semantic category, and calendar features provide complementary information.

The findings support the value of semantics-aware decision layers in smart-city forecasting, especially when service categories differ in temporal behavior and operational trade-offs. Future work should test transferability, deepen fairness analysis of reporting bias, compare abnormal-demand definitions, and link alerts to operational response.

Declarations

Data and code availability. The study uses public NYC 311, NOAA, OpenStreetMap, and PLUTO-derived inputs; processed data and scripts can be shared subject to source licensing, redistribution, and storage constraints.

Competing interests. The authors declare no competing interests.

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Author contributions. Tran Dai Phong Lam led data processing, modeling, analysis, and drafting; Thu Le supervised study design and editing; Nguyen Quoc Hung and Nguyen Trung Trinh contributed to validation, experiment review, and editing.

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